

Neutrino Oscillation as the Unique Admissible Reconstruction of Flavor Under Noncommuting Propagation

Identity Non-Conservation Under Admissible Propagation

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AASC universal-obstruction mathematical-physics version

Abstract

Neutrino oscillation is ordinarily described as the consequence of mass–flavor misalignment, phase accumulation, and projection back onto weak-interaction flavor states. That account is experimentally successful and is not modified here. The present paper proves a narrower closure result: once flavor is fixed only at weak-interaction boundary events, and propagation is governed by an operator that does not preserve the flavor projectors, globally transported weak-flavor identity is not an admissible same-domain invariant.

This universal-obstruction version makes the mathematical comparison class explicit. It introduces the finite PMNS reconstruction category $\mathcal{C}_{\text{PMNS}}$, whose objects are coherent source–propagation–detection systems $(\mathcal{H}, \{P_\alpha\}, \mathcal{U}(t), B_{\text{in}}, B_{\text{out}}, \mathcal{W})$, and whose morphisms preserve boundary projectors, transition witnesses, coherent interference terms, basis covariance, no-selector discipline, and no-repair history. Within this category, uniqueness is not raw metaphysical uniqueness; it is equivalence-class uniqueness under witness-preserving bookkeeping isomorphism. This version also makes literal the categorical language: identity morphisms, composition, quotienting by admissible equivalence, and the reconstruction functor to transition witnesses are stated explicitly. It then completes the partial-algebra obstruction layer: globally transported flavor is treated as an attempted totalization of a boundary-defined partial operator inside a partial reconstruction category $\mathcal{C}_{\text{PMNS}}^\partial$, and hidden transported-flavor extensions collapse by conservative reconstruction equivalence or fail one of the exhaustion classes. The final quotient $\mathcal{Q}_{\text{PMNS}}^\partial$ is universal for admissible PMNS reconstruction: every admissible reconstruction factors through it uniquely up to witness-preserving bookkeeping equivalence.

A witness-completion theorem then derives the declared completeness of weak-interaction flavor witnesses from finite PMNS scope, detector coupling structure, and admissible observability. Any additional flavor witness either belongs to the boundary algebra generated by the weak flavor projectors, changes detector transition statistics, remains empirically disconnected, or exits the declared PMNS scope. Finally, an AQFT embedding reads the same argument through localized boundary algebras and reconstruction sectors: flavor witnesses live at source and detector regions, while coherent propagation supplies the interior evolution but no propagation-local flavor selector.

AASC is used throughout as a constraint formalism, not as additional neutrino dynamics. Its kernel–admissibility, standing, reference, and irreversibility—is instantiated whenever the source-propagation-detection history is treated as a coherent construction of an evaluable transition witness. The resulting theorem is conditional and scope-locked: within $\mathcal{C}_{\text{PMNS}}$, coherent noncommuting propagation excludes globally transported weak-flavor identity, and the standard oscillatory boundary-to-boundary kernel is the unique admissible reconstruction class up to bookkeeping-equivalent redescription.

Main sequent. On the declared neutrino domain,

$$\text{Tr}_E(\alpha), \quad \text{Tr}_D(\beta), \quad [H, P_\alpha] \neq 0, \quad \text{Adm}(\mathcal{W}_{\alpha\beta}) \implies \neg \text{Stand}(\alpha \text{ as interior flavor}) \wedge \text{Stand}(K_{\beta\alpha}), \quad (1)$$

where $K_{\beta\alpha} = \langle \nu_\beta | \mathcal{U} | \nu_\alpha \rangle$ is the standard PMNS boundary amplitude. If the coherent off-diagonal entries are nonzero on the declared interval class, this standing object is observed as oscillation.

A second, prior sequent fixes the AASC load-bearing status used throughout:

$$\text{CC}(\mathcal{D}_\nu) \implies K_{\text{AASC}} = \{\text{Adm}, \text{Stand}, \text{Ref}, \text{lrr}\} \text{ is instantiated on } \mathcal{D}_\nu. \quad (2)$$

Here $\text{CC}(\mathcal{D}_\nu)$ means that the source trace, propagation interval, and detector trace are being evaluated as one coherent construction of a boundary witness. Equation (2) is the reason the closure proof is not merely semantic: it states the minimal constraint package without which the target object is no longer a coherent same-domain construction.

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1 Introduction: the theorem target

The ordinary formalism of neutrino oscillation is not in dispute. A neutrino produced in association with a charged lepton of flavor α is represented at the source by a weak-interaction flavor state $|\nu_\alpha\rangle$. Propagation is diagonal, or approximately diagonal, in another basis. Detection projects the propagated state against weak-interaction flavor states. The resulting probabilities depend on the PMNS matrix, mass-squared differences, phases, baseline, energy, matter effects, and coherence assumptions [1, 2, 3, 4, 5, 6, 7, 8].

The target question is not how to compute those probabilities. It is whether the source flavor label may be treated as a globally carried identity during the propagation interval. Standard language often tolerates both of the following sentences:

- (a) the flavor label is fixed at weak-interaction production and detection events;
- (b) the neutrino has that flavor identity while propagating.

The first sentence is operationally licensed. The second requires a transport invariant. When the propagation operator fails to preserve the flavor projectors, that invariant is absent.

The novelty is therefore not a replacement of PMNS dynamics. The novelty is a closure classification of identity claims. The standard formalism says how the amplitudes evolve. This paper says which identity assignments remain admissible once those amplitudes, boundary labels, and scope assumptions are already fixed. The result is stronger than the familiar statement that flavor and mass bases differ: basis mismatch is a representational fact; noncommuting propagation plus boundary fixation plus a conserved-identity claim is an admissibility obstruction.

Claim. In a fixed neutrino boundary-propagation domain with weak-interaction flavor traces and noncommuting propagation, no same-domain standing-preserving conserved flavor label exists. The unique admissible same-scope reconstruction class, up to bookkeeping-equivalent redescription, is the boundary-to-boundary transition kernel. In

the coherent nontrivial case, that equivalence class is realized empirically as neutrino oscillation.

The AASC upgrade matters at the last sentence. A bare operator argument proves that fixed flavor transport fails. A hostile referee can still ask why oscillation, rather than an arbitrary reinterpretation, hidden transport label, or additional witness sector, is the forced same-scope replacement. The AASC closure architecture supplies the missing exhaustion: no second same-domain classifier, no boundary selector, no present-only identity under irreversible commitment, no same-scope operator rescue, no role beyond anchor/tensor/skin, and no hidden-witness sector inside the unchanged finite PMNS witness scope, and no BSM counterexample without explicit scope change [9, 10, 11, 15, 13, 14, 16, 17].

2 Scope lock and non-claims

The theorem is conditional and fixed-domain. A comparison is same-scope only when it preserves all of the following: the weak-interaction boundary trace maps, the declared propagation Hamiltonian or effective Hamiltonian, the coherent witness family, the boundary output alphabet, the role typing of components, and the envelope ledger of numerical inputs. Changing any of these may be useful physics, but it is not a same-domain rescue of fixed flavor transport.

Definition 2.1 (Same-scope comparison). A proposal R is a same-scope competitor to the present theorem only if it preserves the tuple

$$(\mathcal{H}_\nu, \mathcal{F}, H, \mathcal{U}, \text{Tr}_E, \text{Tr}_D, \mathcal{W}, \mathcal{S}, \mathcal{E}) \quad (3)$$

up to admissible redescription. Here $\mathcal{H}_\nu$ is the neutrino state space, $\mathcal{F}$ is the weak flavor alphabet, H is the declared Hamiltonian or effective Hamiltonian, $\mathcal{U}$ is the propagation family, Tr_E and Tr_D are boundary trace maps, $\mathcal{W}$ is the family of admitted transition observables, $\mathcal{S}$ records approximations and physical regime assumptions, and $\mathcal{E}$ records numerical inputs not derived by the theorem.

Definition 2.2 (Finite witness-complete PMNS scope). A finite witness-complete PMNS scope is a same-scope comparison domain in which: (i) the relevant propagation sector is finite-dimensional; (ii) the source and detector flavor labels are fixed only by weak-interaction boundary traces; (iii) the admitted flavor witnesses are the standard boundary amplitudes or probabilities generated by the declared PMNS-type propagation envelope; (iv) coherence assumptions are fixed in $\mathcal{S}$; and (v) no additional flavor readout, sterile channel, hidden detector, collapse map, or new interaction witness is part of the declared witness family. This is the ordinary finite oscillation reconstruction setting, not a claim about all possible larger-domain neutrino physics.

Theorem 2.3 (Finite PMNS-scope closure theorem). *Within a finite witness-complete PMNS scope, a proposed reconstruction of weak flavor is a same-scope competitor only if it preserves the boundary flavor traces, the declared propagation tensor, the admitted transition witnesses, basis covariance, no-selector discipline, and no-repair discipline. A proposal that adds a new flavor witness sector, extra detector, hidden branch, sterile channel, collapse rule, or new interaction channel is not a same-scope fixed-flavor rescue; it is either bookkeeping-equivalent to the boundary kernel on the declared witnesses or a larger-domain proposal.*

Proof. This is the finite-PMNS specialization of fixed-domain exhaustion and same-scope operator closure. In the declared scope the available admissibility-bearing loci are the boundary anchors,

propagation tensor, admitted witness family, skin/covariance structure, no-selector boundary, and no-repair history. A proposal that changes none of these induces the same transition witness and is bookkeeping-equivalent. A proposal that changes one of them exits the declared comparison class. Since the finite PMNS scope expressly contains no further admissible flavor witness sector, no residual same-scope locus remains for a conserved weak-flavor carrier. $\square$

The phrase “oscillation is unique” means equivalence-class uniqueness inside this finite, witness-complete PMNS comparison class. If the comparison class is widened by adding sterile sectors, extra detection channels, collapse dynamics, decoherence maps, nonstandard interactions, or new envelope data, the theorem is not falsified; the physical scope has changed and must be evaluated under a new tuple.

The present result is conditional in the standard mathematical sense: if coherent witness-preserving reconstruction is required for the source-propagation-detection construction, then globally transported weak-flavor identity is excluded in the declared scope.

The result does not claim any of the following:

- it does not derive the PMNS matrix, mixing angles, mass-squared splittings, CP phase, absolute mass scale, or matter potential;
- it does not modify quantum field theory, the Standard Model, or the ordinary Hilbert-space calculation;
- it does not deny neutrino masses or the empirical reality of oscillation;
- it does not deny flavor as an interaction label;
- it does not claim that no larger-domain or BSM model can be investigated;
- it does not treat unitary propagation as thermodynamically irreversible;
- it does not reduce physical explanation to semantics.

Remark 2.4 (Irreversibility). Irreversibility is used here in the AASC finite-history sense. A source trace, propagation interval, and detector trace form a committed evaluated history. A later description may not rewrite an earlier endpoint label, add an undeclared selector, or change the target domain while claiming to preserve the same evaluated act. This is compatible with unitary invertibility of $\mathcal{U}$.

Remark 2.5 (Anti-realism insulation). The analysis does not deny the reality of the propagation structure or the observed oscillation phenomenon. It denies only the stronger assertion that weak flavor is a propagation-invariant identity assignment in a domain where the flavor projectors are not invariant.

3 Orthodox PMNS interface

Let $\mathcal{H}_\nu$ be the one-particle neutrino Hilbert space in the approximation under discussion. The weak-interaction flavor states and propagation mass states are related by

$$|\nu_\alpha\rangle = \sum_i U_{\alpha i}^* |\nu_i\rangle, \quad \alpha \in \{e, \mu, \tau\}, \quad (4)$$

where U is the PMNS matrix. In vacuum, suppressing wave-packet details, propagation is diagonal in the mass basis:

$$\mathcal{U}(L, E)|\nu_i\rangle = e^{-im_i^2 L/(2E)}|\nu_i\rangle. \quad (5)$$

Equivalently, in the mass basis,

$$H_{\text{vac}} \simeq \frac{1}{2E} \text{diag}(m_1^2, m_2^2, m_3^2), \quad \mathcal{U}(L, E) = e^{-iH_{\text{vac}}L}. \quad (6)$$

Matter effects replace H_{vac} by the declared effective Hamiltonian. Nothing in the closure proof depends on using the vacuum Hamiltonian rather than the effective Hamiltonian; it depends only on whether the flavor projectors commute with the declared propagation operator.

Define flavor projectors

$$P_\alpha = |\nu_\alpha\rangle\langle\nu_\alpha|. \quad (7)$$

The standard boundary-to-boundary amplitude is

$$K_{\beta\alpha}(L, E) = \langle\nu_\beta|\mathcal{U}(L, E)|\nu_\alpha\rangle = \sum_i U_{\beta i} e^{-im_i^2 L/(2E)} U_{\alpha i}^*, \quad (8)$$

with probability

$$P_{\alpha\rightarrow\beta}(L, E) = |K_{\beta\alpha}(L, E)|^2. \quad (9)$$

A fixed flavor identity would require more than nonzero transition amplitudes. It would require invariance of the flavor projector under propagation:

$$\mathcal{U}(L)^\dagger P_\alpha \mathcal{U}(L) = P_\alpha \quad \text{for all licensed intervals } L. \quad (10)$$

When this condition fails, flavor remains meaningful as a boundary trace but not as a transported interior identity.

4 The Operator Category of PMNS Reconstructions

This section fixes the exact comparison class over which the uniqueness and exhaustion claims range. The category is intentionally finite and scope-locked. It formalizes the ordinary coherent PMNS reconstruction problem rather than a full quantum-field-theoretic model of neutrino production and detection.

Definition 4.1 (Finite PMNS reconstruction object). Let $\mathcal{C}_{\text{PMNS}}$ be the category whose objects are finite coherent PMNS reconstruction systems

$$\mathfrak{R} = (\mathcal{H}, \{P_\alpha\}_{\alpha \in \mathcal{F}}, \mathcal{U}(t), B_{\text{in}}, B_{\text{out}}, \mathcal{W}), \quad (11)$$

where $\mathcal{H}$ is the finite-dimensional neutrino propagation Hilbert space in the declared approximation; $\mathcal{F} = \{e, \mu, \tau\}$, or $\mathcal{F} = \{e, \mu\}$ in the two-flavor reduction; P_α are weak-interaction flavor projectors at boundary events; $\mathcal{U}(t) = e^{-iHt}$ is coherent propagation generated by the declared Hamiltonian or effective Hamiltonian; B_{in} and B_{out} are the source and detector boundary interfaces; and $\mathcal{W}$ is the witness map extracting transition data. For a density matrix ρ prepared through boundary channel α , the witness map may be written

$$\mathcal{W}(\alpha, t, \beta) = \text{Tr}\left(P_\beta \mathcal{U}(t) P_\alpha \rho P_\alpha \mathcal{U}(t)^\dagger\right), \quad (12)$$

and, for pure boundary states,

$$\mathcal{W}_{\alpha\rightarrow\beta}(t) = |\langle\nu_\beta|\mathcal{U}(t)|\nu_\alpha\rangle|^2. \quad (13)$$

Definition 4.2 (Admissible reconstruction morphism). A morphism $\Phi : \mathfrak{R} \rightarrow \mathfrak{R}'$ in $\mathcal{C}_{\text{PMNS}}$ is an admissible reconstruction-preserving map satisfying the following five conditions.

(M1) **Boundary-projector preservation.** There is a boundary-label relabeling $\sigma : \mathcal{F} \rightarrow \mathcal{F}'$ such that

$$\Phi(P_\alpha) = P'_{\sigma(\alpha)}. \quad (14)$$

(M2) **Transition-witness preservation.** For all admitted boundary labels and times,

$$\mathcal{W}'_{\sigma(\alpha) \rightarrow \sigma(\beta)}(t) = \mathcal{W}_{\alpha \rightarrow \beta}(t). \quad (15)$$

(M3) **Interference preservation.** The coherent cross terms in the amplitude decomposition are preserved up to unitary bookkeeping equivalence. A morphism may change charts or phases, but may not replace a coherent amplitude witness by an incoherent mixture while remaining in the same coherent object class.

(M4) **Basis covariance.** For any propagation-basis change V , the simultaneous transformation

$$\mathcal{U}(t) \mapsto V\mathcal{U}(t)V^\dagger, \quad P_\alpha \mapsto VP_\alpha V^\dagger \quad (16)$$

does not alter the witness class.

(M5) **No-selector condition.** No morphism may introduce an additional map

$$S : \mathcal{H} \rightarrow \mathcal{F} \quad (17)$$

assigning definite intermediate flavor labels unless S is definable from the boundary witness algebra and preserves the transition witness.

Definition 4.3 (Admissible reconstruction equivalence). Two reconstruction objects $\mathfrak{R}, \mathfrak{R}' \in \text{Obj}(\mathcal{C}_{\text{PMNS}})$ are admissibly equivalent, written

$$\mathfrak{R} \sim_{\text{adm}} \mathfrak{R}', \quad (18)$$

iff there exist admissible reconstruction morphisms both ways preserving the boundary witnesses and interference structure. The uniqueness claim in this paper is uniqueness of the equivalence class $[\mathfrak{R}]_{\text{adm}}$, not uniqueness of a preferred notation, basis, chart, or verbal interpretation. The no-discrete-multiplicity and admissible-interior uniqueness principles used here are the finite PMNS instance of the apex closure package [23].

Proposition 4.4 ($\mathcal{C}_{\text{PMNS}}$ is a category). *The finite PMNS reconstruction objects together with admissible reconstruction morphisms form a category. For every object $\mathfrak{R}$, the identity morphism $\text{id}_{\mathfrak{R}}$ is admissible. If $\Phi : \mathfrak{R} \rightarrow \mathfrak{R}'$ and $\Psi : \mathfrak{R}' \rightarrow \mathfrak{R}''$ are admissible morphisms, then $\Psi \circ \Phi : \mathfrak{R} \rightarrow \mathfrak{R}''$ is admissible. Composition is associative because the underlying Hilbert-space maps, boundary relabelings, and witness-preserving transformations compose associatively.*

Proof. The identity map sends each boundary projector to itself, preserves every transition witness, preserves each coherent cross term, is covariant under a simultaneous basis change, and introduces no selector. Hence $\text{id}_{\mathfrak{R}}$ is admissible. For composition, suppose Φ has label relabeling σ and Ψ has label relabeling τ . Then $\Psi \circ \Phi$ has relabeling $\tau \circ \sigma$, so boundary projectors are preserved. Transition witnesses are preserved because

$$\mathcal{W}''_{(\tau \circ \sigma)(\alpha) \rightarrow (\tau \circ \sigma)(\beta)}(t) = \mathcal{W}'_{\sigma(\alpha) \rightarrow \sigma(\beta)}(t) = \mathcal{W}_{\alpha \rightarrow \beta}(t). \quad (19)$$

Interference preservation and basis covariance are stable under composition of unitary bookkeeping transformations. A selector cannot appear under composition unless it appears in one of the two factors, which admissibility forbids. Thus admissible morphisms compose, and ordinary associativity of function/operator composition gives associativity. $\square$

Definition 4.5 (Admissible quotient category). The admissible quotient category $\mathcal{C}_{\text{PMNS}}/\sim_{\text{adm}}$ has objects the equivalence classes $[\mathfrak{R}]_{\text{adm}}$ and morphisms induced by admissible reconstruction morphisms modulo witness-preserving bookkeeping equivalence. The main uniqueness result is a statement in $\mathcal{C}_{\text{PMNS}}/\sim_{\text{adm}}$: there is one nontrivial coherent oscillatory reconstruction class for the declared finite PMNS witness scope.

Definition 4.6 (Transition-witness category and reconstruction functor). Let $\mathcal{W}_{\text{PMNS}}$ be the category whose objects are transition-witness families $\{\mathcal{W}_{\alpha \rightarrow \beta}(t)\}$, with morphisms given by boundary-label relabelings and unitary chart changes preserving all transition probabilities and coherent interference terms. Define the reconstruction functor

$$\mathfrak{F} : \mathcal{C}_{\text{PMNS}} \longrightarrow \mathcal{W}_{\text{PMNS}} \quad (20)$$

by

$$\mathfrak{F}(\mathfrak{R}) = \{\mathcal{W}_{\alpha \rightarrow \beta}(t)\}_{\alpha, \beta, t}, \quad \mathfrak{F}(\Phi) = \text{the induced relabeling/chart map on witnesses.} \quad (21)$$

Proposition 4.7 (Functoriality of witness reconstruction). *The assignment $\mathfrak{F} : \mathcal{C}_{\text{PMNS}} \rightarrow \mathcal{W}_{\text{PMNS}}$ is a functor. It sends identities to identities and composition of admissible reconstruction morphisms to composition of witness-preserving maps.*

Proof. For any reconstruction object, $\mathfrak{F}(\text{id}_{\mathfrak{R}})$ is the identity map on the witness family. If Φ and Ψ are composable admissible morphisms with relabelings σ and τ , the induced map on witness families has relabeling $\tau \circ \sigma$, which is exactly the composition of the induced witness maps. Because admissible morphisms preserve transition witnesses and coherent interference terms, the image lies in $\mathcal{W}_{\text{PMNS}}$. Therefore $\mathfrak{F}$ is functorial. $\square$

Corollary 4.8 (Uniqueness as admissible preimage uniqueness). *The oscillation uniqueness claim may be expressed as admissible preimage uniqueness: within the declared finite PMNS scope, the witness family $\mathfrak{F}(\mathfrak{R})$ has, up to $\sim_{\text{adm}}$, a single nontrivial coherent reconstruction preimage in $\mathcal{C}_{\text{PMNS}}$. Any alleged second preimage either is isomorphic in $\mathcal{C}_{\text{PMNS}}/\sim_{\text{adm}}$, violates a morphism condition, or changes the object class by enlarging the detector algebra, propagation law, or admitted witness family.*

Definition 4.9 (Partial PMNS reconstruction object). The partial PMNS reconstruction category $\mathcal{C}_{\text{PMNS}}^{\partial}$ refines $\mathcal{C}_{\text{PMNS}}$ by making the domain of weak-flavor transport intrinsic to each object. An object is a tuple

$$\mathfrak{R}^{\partial} = (\mathfrak{R}, \{\mathcal{T}_{\alpha} : X_{\nu} \rightharpoonup \mathcal{F}\}_{\alpha \in \mathcal{F}}, \mathcal{E}_{\partial}, \mathcal{E}_{\text{prop}}), \quad (22)$$

where $\mathfrak{R} \in \text{Obj}(\mathcal{C}_{\text{PMNS}})$, X_{ν} is the source–propagation–detection carrier, $\mathcal{E}_{\partial} = \mathcal{E}_{\text{in}} \cup \mathcal{E}_{\text{out}}$ is the boundary witness envelope, and $\mathcal{E}_{\text{prop}}$ is the propagation interior. For each α , the partial flavor transport operator satisfies

$$\text{dom}(\mathcal{T}_{\alpha}) = \mathcal{E}_{\partial}, \quad \mathcal{E}_{\text{prop}} \cap \text{dom}(\mathcal{T}_{\alpha}) = \emptyset. \quad (23)$$

Thus weak flavor is a boundary-defined operator, while the propagation interior is represented by $\mathcal{U}(t)$ and the boundary-to-boundary kernel, not by a transported weak-flavor selector.

Definition 4.10 (Domain-preserving partial reconstruction morphism). A morphism $\Phi^\partial : \mathfrak{R}^\partial \rightarrow \mathfrak{R}'^\partial$ in $\mathcal{C}_{\text{PMNS}}^\partial$ is an admissible reconstruction morphism $\Phi : \mathfrak{R} \rightarrow \mathfrak{R}'$ in $\mathcal{C}_{\text{PMNS}}$ together with preservation of partial-operator domains:

$$\Phi(\mathcal{E}_\partial) \subseteq \mathcal{E}'_\partial, \quad \Phi(\mathcal{E}_{\text{prop}}) \subseteq \mathcal{E}'_{\text{prop}}, \quad \Phi(\text{dom}(\mathcal{T}_\alpha)) \subseteq \text{dom}(\mathcal{T}'_{\sigma(\alpha)}). \quad (24)$$

No morphism in $\mathcal{C}_{\text{PMNS}}^\partial$ may send a point of the old propagation interior to a defined weak-flavor value unless that point is first retyped as boundary data in a new physical scope. This is the categorical form of operator-domain fixity.

Proposition 4.11 ($\mathcal{C}_{\text{PMNS}}^\partial$ is a category). *Partial PMNS reconstruction objects and domain-preserving partial reconstruction morphisms form a category. The identity morphism preserves $\mathcal{E}_\partial$, $\mathcal{E}_{\text{prop}}$, and every partial domain. Composition preserves these inclusions and therefore cannot create a totalized flavor operator on the old propagation carrier.*

Proof. The identity morphism is admissible in $\mathcal{C}_{\text{PMNS}}$ by Proposition 4.4 and fixes $\mathcal{E}_\partial$, $\mathcal{E}_{\text{prop}}$, and $\text{dom}(\mathcal{T}_\alpha)$. If Φ^∂ and Ψ^∂ are composable, their underlying maps compose in $\mathcal{C}_{\text{PMNS}}$. Domain preservation is stable under composition because

$$(\Psi \circ \Phi)(\mathcal{E}_\partial) \subseteq \Psi(\mathcal{E}'_\partial) \subseteq \mathcal{E}''_\partial, \quad (\Psi \circ \Phi)(\mathcal{E}_{\text{prop}}) \subseteq \Psi(\mathcal{E}'_{\text{prop}}) \subseteq \mathcal{E}''_{\text{prop}}, \quad (25)$$

and the same inclusion holds for the domains of $\mathcal{T}_\alpha$. Associativity is inherited from composition in $\mathcal{C}_{\text{PMNS}}$. Hence $\mathcal{C}_{\text{PMNS}}^\partial$ is a category. $\square$

Definition 4.12 (Universal admissible reconstruction quotient). Let

$$\pi : \mathcal{C}_{\text{PMNS}}^\partial \longrightarrow \mathcal{Q}_{\text{PMNS}}^\partial \quad (26)$$

be the quotient functor identifying two partial PMNS reconstruction objects exactly when they are related by admissible domain-preserving morphisms both ways and preserve all boundary transition witnesses. The quotient $\mathcal{Q}_{\text{PMNS}}^\partial$ is the admissible reconstruction quotient of the declared finite PMNS scope.

Proposition 4.13 (Categorical quotient universal property). *Let $\pi := \pi : \mathcal{C}_{\text{PMNS}}^\partial \rightarrow \mathcal{Q}_{\text{PMNS}}^\partial$ be the quotient functor. For any category $\mathcal{D}$ and any functor $G : \mathcal{C}_{\text{PMNS}}^\partial \rightarrow \mathcal{D}$ that is constant on admissible-equivalence classes of objects and morphisms, there exists a unique functor $\overline{G} : \mathcal{Q}_{\text{PMNS}}^\partial \rightarrow \mathcal{D}$ such that*

$$G = \overline{G} \circ \pi. \quad (27)$$

Thus $\mathcal{Q}_{\text{PMNS}}^\partial$ is the categorical quotient of $\mathcal{C}_{\text{PMNS}}^\partial$ by witness-preserving admissible equivalence.

Proof. Set $\overline{G}([X]_{\text{adm}}) = G(X)$ and similarly on morphism classes. This is well-defined precisely because G is constant on admissible-equivalence classes. The identity $G = \overline{G} \circ \pi$ is immediate, and uniqueness follows from surjectivity of π onto quotient object and morphism classes. $\square$

Theorem 4.14 (Universal factorization through the admissible quotient). *Let $\mathcal{D}$ be any category of same-scope PMNS reconstructions whose morphisms preserve boundary witnesses, coherent interference terms, basis covariance, no-selector discipline, and partial-domain fixity. If $F : \mathcal{C}_{\text{PMNS}}^\partial \rightarrow \mathcal{D}$ sends admissibly equivalent partial reconstructions to the same object of $\mathcal{D}$, then F factors uniquely through $\mathcal{Q}_{\text{PMNS}}^\partial$:*

$$F = \overline{F} \circ \pi \quad (28)$$

for a unique functor $\bar{F} : \mathcal{Q}_{\text{PMNS}}^\partial \rightarrow \mathcal{D}$. Consequently, every admissible same-scope PMNS reconstruction factors through the universal admissible quotient, and all remaining variation is bookkeeping equivalence.

Proof. Define $\bar{F}([\mathfrak{R}^\partial]_{\text{adm}}) = F(\mathfrak{R}^\partial)$. This is well-defined because F identifies admissibly equivalent partial reconstructions. Define $\bar{F}$ on morphism classes by applying F to any representative morphism. Again this is well-defined because morphisms differing only by witness-preserving bookkeeping equivalence have the same image under F . Then $F = \bar{F} \circ \pi$ by construction. Uniqueness follows because π is surjective on quotient objects and morphism classes. Thus $\mathcal{Q}_{\text{PMNS}}^\partial$ is universal for admissible same-scope PMNS reconstruction. $\square$

Corollary 4.15 (Universal quotient form of oscillation uniqueness). *In $\mathcal{Q}_{\text{PMNS}}^\partial$, the finite witness-complete noncommuting PMNS domain has a single nontrivial coherent reconstruction object represented by the boundary kernel $K_{\beta\alpha} = \langle \nu_\beta | \mathcal{U} | \nu_\alpha \rangle$. Any alleged second object either maps to the same quotient object, violates partial-domain preservation, or changes physical scope.*

Definition 4.16 (Admissible reconstruction rule). Let $\mathfrak{R}(\mathcal{D}_\nu)$ be the class of rules represented by objects and morphisms in $\mathcal{C}_{\text{PMNS}}$, assigning source flavor traces, the declared propagation family, detector traces, and envelope data to admitted transition amplitudes or probabilities. A rule $R \in \mathfrak{R}(\mathcal{D}_\nu)$ is same-scope admissible only if it preserves boundary anchors, propagation tensor, transition witnesses, coherent interference, covariance under skin, no hidden selector, and no repair of act-time boundary assignments.

Theorem 4.17 (Closure of the PMNS reconstruction category). *Within $\mathcal{C}_{\text{PMNS}}$, every admissible reconstruction preserving coherent transition witnesses falls into exactly one of the following classes: (i) the standard oscillatory boundary-kernel reconstruction class; (ii) a boundary-label relabeling or unitary chart redescription of that class; or (iii) a non-admissible or larger-scope object outside $\mathcal{C}_{\text{PMNS}}$.*

Proof. Let $R \in \mathfrak{R}(\mathcal{D}_\nu)$ be a same-scope reconstruction rule. A fixed flavor transport account requires $[P_\alpha, \mathcal{U}(t)] = 0$ for the relevant boundary projector and all admitted propagation intervals. In the nontrivial PMNS case this fails, so fixed flavor transport cannot be an object-preserving reconstruction. A hidden selector account introduces $S : \mathcal{H} \rightarrow \mathcal{F}$. If S affects detection statistics, condition (M2) is violated. If it preserves all statistics, it is bookkeeping-inert. If it is not boundary-witness-definable, condition (M5) is violated. A basis relabeling is admissible only when it satisfies (M1)–(M4), in which case it is an isomorphism in $\mathcal{C}_{\text{PMNS}}$, not a rival reconstruction. Finally, adding new observables, detector channels, sterile sectors, collapse maps, or empirically coupled hidden witnesses changes the object class and exits $\mathcal{C}_{\text{PMNS}}$. These cases exhaust the morphism-classified alternatives. Proposition 4.4 ensures that this is a literal categorical comparison class, and Corollary 4.8 expresses the same closure as uniqueness of admissible preimages under the reconstruction functor. $\square$

Theorem 4.18 (Categorical reconstruction exhaustion). *Within $\mathcal{C}_{\text{PMNS}}$ on a finite witness-complete PMNS scope, every same-scope admissible reconstruction rule $R \in \mathfrak{R}(\mathcal{D}_\nu)$ is bookkeeping-equivalent to the boundary kernel*

$$K_{\beta\alpha}(t) = \langle \nu_\beta | \mathcal{U}(t) | \nu_\alpha \rangle. \quad (29)$$

Conversely, any reconstruction not bookkeeping-equivalent to K violates boundary-projector preservation, transition-witness preservation, interference preservation, basis covariance, no-selector discipline, no-repair discipline, or the fixed scope itself.

Proof. A reconstruction rule satisfying boundary-projector preservation receives its source and detector flavor references from P_α and P_β . A rule satisfying tensor preservation uses the same propagation family $\mathcal{U}(t)$. Their amplitude-level composition is therefore the matrix element $\langle \nu_\beta | \mathcal{U}(t) | \nu_\alpha \rangle$. Transition-witness preservation fixes its admitted probability witness; interference preservation prevents replacement by a classical mixture; basis covariance identifies unitary chart variants; the no-selector condition forbids an intermediate flavor representative not fixed by the boundary algebra; no-repair forbids detector-side reassignment of the source event. Thus every admissible member of $\mathcal{C}_{\text{PMNS}}$ induces the same witness class as K . A non-equivalent rule must differ at one of the morphism-protected loci; hence it either violates admissibility or changes scope. $\square$

5 Witness Completion for Finite PMNS Flavor Scope

The preceding category still leaves one possible attack: perhaps weak-interaction boundary projectors do not exhaust the flavor witnesses. The following theorem derives the declared witness completeness from the finite PMNS scope, detector coupling structure, and admissible observability.

Definition 5.1 (Detector-coupled finite PMNS scope). A detector-coupled finite PMNS scope is

$$\mathcal{S}_{\text{PMNS}} = (\mathcal{H}_{\text{PMNS}}, \mathcal{U}(t), \{P_\alpha\}_{\alpha \in \mathcal{F}}, \mathcal{D}), \quad (30)$$

where $\mathcal{D}$ is the class of detector couplings that operationally fix flavor through charged-current weak interaction channels

$$\nu_\alpha + N \longrightarrow \ell_\alpha + X. \quad (31)$$

A flavor witness is admissible in this scope only when it is recoverable from detector-accessible weak-interaction events.

Definition 5.2 (Boundary flavor witness algebra). The boundary flavor witness algebra is

$$\mathcal{A}_\partial = \text{Alg}\{P_e, P_\mu, P_\tau\}, \quad (32)$$

or the corresponding two-flavor subalgebra in a reduced example. It is the algebra generated by the charged-current flavor projectors available to the declared detector-coupling class.

Definition 5.3 (Minimal admissible flavor witness algebra). A boundary witness algebra $\mathcal{B}$ is admissibly complete for finite PMNS flavor if every detector-accessible flavor readout factors through $\mathcal{B}$ and the declared transition kernel. It is minimal if every proper subalgebra fails to distinguish at least one admitted charged-current flavor boundary channel.

Definition 5.4 (Witness-complete flavor scope). A neutrino domain is witness-complete for weak flavor when every admissible flavor readout in the declared finite PMNS scope is generated by $\mathcal{A}_\partial$ together with the declared transition kernel. This is a scope theorem, not a metaphysical completeness claim about all possible neutrino physics.

Theorem 5.5 (Minimality of the boundary flavor witness algebra). *Within detector-coupled finite PMNS scope, $\mathcal{A}_\partial = \text{Alg}\{P_e, P_\mu, P_\tau\}$ is the minimal admissibly complete flavor witness algebra. It is complete because every admitted flavor readout is mediated by the declared charged-current detector couplings; it is minimal because removing any boundary projector removes the corresponding charged-current flavor boundary channel from the witness alphabet.*

Proof. Completeness follows from the declared detector-coupling class: the admitted flavor outcomes are precisely the charged-current lepton channels, and these channels are represented by the projectors generating $\mathcal{A}_\partial$. Any admitted transition witness is formed by preparing a boundary channel, propagating coherently, and reading out another boundary channel; hence it factors through $\mathcal{A}_\partial$ and the transition kernel. Minimality is equally direct. If a projector P_α is removed from the generating set, the boundary event in which the charged lepton ℓ_α fixes flavor is no longer represented in the algebra. The reduced algebra therefore fails to preserve the declared detector alphabet and is not an admissibly complete flavor witness algebra for the same PMNS scope. $\square$

Theorem 5.6 (Weak-interaction flavor witness completion). *Within detector-coupled finite PMNS scope, the weak-interaction boundary projectors $\{P_\alpha\}_{\alpha \in \mathcal{F}}$ form a complete admissible flavor witness family. Equivalently, any proposed flavor witness Q falls into exactly one of four classes: (i) $Q \in \mathcal{A}_\partial$ or is generated by $\mathcal{A}_\partial$ and the transition kernel; (ii) Q changes detector transition statistics; (iii) Q is empirically disconnected from flavor detection; or (iv) Q extends the physical scope beyond finite PMNS reconstruction.*

Proof. Flavor observability in the declared oscillation scope is detector-mediated: a flavor label is fixed through boundary charged-current events, not through an independently measured propagation observable. Thus the detector channels generate $\mathcal{A}_\partial$, and Theorem 5.5 identifies this algebra as minimal and admissibly complete for the declared detector alphabet. Let Q be an alleged additional flavor witness. If Q is measurable through the declared charged-current detector couplings without changing any transition witness, then Q is already represented in $\mathcal{A}_\partial$ or in the boundary-kernel data generated from it. If Q couples to detector outcomes in a new way, then the transition statistics or event alphabet changes and the declared finite PMNS scope has been enlarged. If Q does not couple to admitted detector outcomes, it is not a flavor witness in the admissible observational sense. If Q determines intermediate flavor without boundary fixation, it functions as a hidden selector and violates the no-selector condition in $\mathcal{C}_{\text{PMNS}}$. Therefore $\mathcal{A}_\partial$ is witness-complete within the declared scope. $\square$

Corollary 5.7 (Hidden witness sector collapse). *Any proposed hidden flavor witness sector inside finite PMNS scope satisfies*

$$\begin{aligned} &\text{observable scope extension} \quad \vee \quad \text{bookkeeping redundancy} \\ &\vee \quad \text{empirical disconnection} \quad \vee \quad \text{inadmissible selector.} \end{aligned} \tag{33}$$

Consequently no hidden witness sector supplies a same-scope globally transported weak-flavor identity.

Proof. If the sector couples observably, it changes $\mathcal{D}$, $\mathcal{W}$, or $\mathcal{S}$. If it induces the same transition witnesses, it is bookkeeping redundancy. If it is disconnected from admitted detector couplings, it is outside the reconstruction problem. If it chooses an intermediate flavor representative without being boundary-witness-definable, it is the hidden selector excluded by $\mathcal{C}_{\text{PMNS}}$ and by the AASC AMetric boundary. These alternatives exhaust the role options available to a proposed witness sector. $\square$

All exhaustion claims below are internal to $\mathcal{C}_{\text{PMNS}}$. Enlarging the detector algebra, adding new interaction channels, or introducing additional empirically coupled sectors defines a different physical scope and is not ruled out by the present theorem.

6 Operator-Domain Obstruction: Flavor Transport as a Partial Operator

The category-theoretic comparison above can be sharpened further using the partial-algebra and operator-domain closure results [20, 21]. The attempted globally transported flavor assignment is not merely a rival interpretation. It is an attempted totalization of a partial operator whose admissible domain is fixed by boundary witness structure.

Definition 6.1 (Flavor transport partial operator). For each weak flavor label $\alpha \in \mathcal{F}$, define the flavor-transport assignment as a partial operator

$$\mathcal{T}_\alpha : X_\nu \rightharpoonup \mathcal{F}, \quad (34)$$

where X_ν is the neutrino source–propagation–detection carrier in the declared finite PMNS scope. Its domain is not the full propagation carrier. It is the union of admissible weak-interaction boundary witness envelopes:

$$\text{dom}(\mathcal{T}_\alpha) = \mathcal{E}_\partial := \mathcal{E}_{\text{in}} \cup \mathcal{E}_{\text{out}}, \quad (35)$$

where $\mathcal{E}_{\text{in}}$ and $\mathcal{E}_{\text{out}}$ are the source and detector boundary envelopes generated by charged-current flavor projectors. Thus $\mathcal{T}_\alpha$ is defined where flavor is fixed by boundary witness coupling. It is undefined on the propagation interior $\mathcal{E}_{\text{prop}}$ except through the boundary-to-boundary reconstruction kernel.

Remark 6.2 (Undefined is not an error state). The undefinedness of $\mathcal{T}_\alpha$ on $\mathcal{E}_{\text{prop}}$ is not empirical ignorance and not a third flavor value. It is non-membership in the admissible operator domain. The interior still carries coherent propagation $\mathcal{U}(t)$; what is absent is an admissibly fixed weak-flavor selector on that interior.

Theorem 6.3 (Flavor transport totalization obstruction). *No admissible extension of the declared finite PMNS reconstruction object totalizes $\mathcal{T}_\alpha$ on the propagation interior. Equivalently, there is no admissible same-scope extension in which boundary-fixed weak flavor becomes a globally defined propagation-invariant assignment on X_ν .*

Proof. Treat the PMNS reconstruction object as a non-degenerate partial-algebraic envelope whose operators include boundary witness formation, coherent propagation, and boundary readout. By Definition 6.1, $\mathcal{T}_\alpha$ is partial: it is defined on $\mathcal{E}_\partial$ and undefined on $\mathcal{E}_{\text{prop}}$. A same-scope globally transported flavor account would require an admissible extension $\mathcal{T}_\alpha^*$ with

$$\text{dom}(\mathcal{T}_\alpha^*) \cap X_\nu = X_\nu, \quad (36)$$

while agreeing with $\mathcal{T}_\alpha$ on the old boundary domain. This is exactly totalization of a partial operator on its original carrier. AASC partial-algebra closure forbids such totalization: admissible extensions preserve old operator domains on the original carrier, since making an undefined old-scope operator application defined is a repair/generator of admissibility rather than an extension of the same envelope. Therefore $\mathcal{T}_\alpha$ cannot be totalized on $\mathcal{E}_{\text{prop}}$ without changing scope or violating admissible domain fixity. $\square$

Corollary 6.4 (No globally transported weak-flavor operator). *A globally transported weak-flavor identity is unavailable in the declared PMNS scope. Any operator purporting to assign weak flavor throughout propagation is either a new-scope operator, a bookkeeping copy of boundary reconstruction data, or an inadmissible totalization of $\mathcal{T}_\alpha$.*

Definition 6.5 (Internal admissible-totalization obstruction object). For a partial PMNS reconstruction object $\mathfrak{R}^\partial$, define the internal admissible-totalization obstruction object

$$\mathfrak{O}_{\text{flavor}}(\mathfrak{R}^\partial) \in \text{Ob}_\partial(\mathfrak{R}^\partial) \quad (37)$$

as the internal obstruction, in the declared admissible reconstruction category, to the attempted totalization

$$\mathcal{T}_\alpha : \mathcal{E}_\partial \rightarrow \mathcal{F} \quad \rightsquigarrow \quad \mathcal{T}_\alpha^* : X_\nu \rightarrow \mathcal{F}. \quad (38)$$

This is not asserted to be a cohomological, K-theoretic, or standard operator-algebraic invariant. By definition, $\mathfrak{O}_{\text{flavor}}(\mathfrak{R}^\partial) = 0$ only if there exists an admissible same-scope extension totalizing $\mathcal{T}_\alpha$ on the original propagation carrier while preserving every boundary transition witness. Otherwise $\mathfrak{O}_{\text{flavor}}(\mathfrak{R}^\partial) \neq 0$.

Theorem 6.6 (Nonvanishing of the flavor obstruction). *For every nontrivial finite PMNS reconstruction object in which weak flavor is boundary-fixed and propagation is noncommuting, the internal admissible-totalization obstruction is nonzero:*

$$\mathfrak{O}_{\text{flavor}}(\mathfrak{R}^\partial) \neq 0. \quad (39)$$

Moreover, any attempted vanishing witness for $\mathfrak{O}_{\text{flavor}}$ must instantiate one of the four reconstruction obstruction classes E1–E4.

Proof. If $\mathfrak{O}_{\text{flavor}} = 0$, then an admissible same-scope totalization $\mathcal{T}_\alpha^* : X_\nu \rightarrow \mathcal{F}$ exists. This contradicts Theorem 6.3, which says that admissible extensions preserve partial-operator domains on the original carrier. Therefore $\mathfrak{O}_{\text{flavor}} \neq 0$. The only ways to present a putative totalization are: choose an interior representative not fixed by the boundary witnesses (E1), repair or generate definedness for an old undefined operator application (E2), authorize the hidden flavor value by a derived carrier such as basis convenience, probability, symmetry, or explanation (E3), or change the detector algebra or sector content while calling it the same PMNS object (E4). These are exactly the obstruction classes of Theorem 6.8. $\square$

Theorem 6.7 (Conservative reconstruction collapse for hidden flavor transport). *Let $\mathfrak{R}^*$ be an admissibility-preserving extension of a finite PMNS reconstruction object that adds a hidden transported-flavor symbol or sector while preserving all boundary transition witnesses $\mathcal{W}_{\alpha \rightarrow \beta}(t)$. Then $\mathfrak{R}^*$ is definitional/bookkeeping-equivalent to the original reconstruction on the old PMNS scope. If it is not bookkeeping-equivalent, it changes detector witnesses, introduces a selector, repairs an undefined assignment, imports a carrier, or changes scope.*

Proof. Conservativity on the old PMNS scope means every old boundary transition witness remains unchanged. If the hidden flavor symbol participates in standing-bearing reconstruction claims while adding no new witness, then admissible conservative-extension collapse applies: new standing-bearing symbols must be explicitly definable from the old boundary witness algebra and transition kernel. Hence the hidden transport is bookkeeping. If the symbol is not definable from the old witness structure, then its alleged standing must come from one of four sources: a selector distinguishing admissibly equivalent interiors, a repair/totalization of an undefined flavor assignment, a carrier predicate authorizing hidden identity, or a scope extension adding new detector-coupled structure. These are precisely the obstruction classes in Theorem 6.8. Thus no nontrivial conservative hidden transport survives. $\square$

Theorem 6.8 (Reconstruction obstruction classification). *Every same-scope attempt to escape oscillatory boundary reconstruction falls into at least one of the following obstruction classes:*

<i>Class</i>	<i>Reconstruction escape route</i>
<i>E1</i>	<i>Selector insertion: a hidden or canonical rule selects an intermediate flavor representative not fixed by boundary witnesses.</i>
<i>E2</i>	<i>Reconstruction repair/generator: an undefined interior flavor assignment is made defined after the fact, or a new eligible assignment is generated from an ineligible one.</i>
<i>E3</i>	<i>Carrier laundering: a derived quantity, probability, symmetry, basis convention, or explanatory virtue is treated as primitive authorization for transported flavor.</i>
<i>E4</i>	<i>Scope transport: the detector algebra, interaction channel, sector content, or domain typing is changed while the result is presented as the same PMNS reconstruction problem.</i>

Each class is inadmissible inside $\mathcal{C}_{\text{PMNS}}$. Therefore no fifth same-scope reconstruction escape remains.

Proof. Let an escape attempt produce a reconstruction not equivalent to the boundary kernel while claiming to preserve the old PMNS scope. If it chooses a definite interior flavor representative, then the choice is not supplied by $\mathcal{A}_\partial$ and is E1. If it makes $\mathcal{T}_\alpha$ defined on $\mathcal{E}_{\text{prop}}$, then it is a totalization/repair/generator and is E2. If it authorizes transported flavor by appeal to mass-basis convenience, likelihood, typicality, explanatory compression, symmetry, or any other derived criterion, then that criterion is functioning as a carrier and is E3. If it adds detector channels, sterile sectors, new couplings, collapse dynamics, or new empirically coupled observables, then the PMNS object has changed and the attempt is E4. These cases exhaust the syntactic routes by which an old-scope reconstruction rule can acquire new admissible coverage: selection, repair/generation, carrier authorization, or scope change. All four are excluded by the AASC operator-exhaustion schema and by the definitions of $\mathcal{C}_{\text{PMNS}}$, $\mathcal{A}_\partial$, and witness-complete PMNS scope. $\square$

Corollary 6.9 (Canonical witness collapse for hidden transported flavor). *Suppose two assignments R_1 and R_2 preserve all PMNS boundary transition witnesses but differ in their hidden transported-flavor reference assignment on the propagation interior. Then the difference is not an admissible same-scope reconstruction distinction. It is either operationally inert bookkeeping or an undefined reference-fixation difference outside the boundary witness algebra.*

Proof. The canonical reference-dependent witness pattern is this: predictive and operational facts remain invariant while reference-dependent ordering or assignment differs [22]. In the neutrino case, the invariant facts are the PMNS transition witnesses $\mathcal{W}_{\alpha \rightarrow \beta}(t)$. If R_1 and R_2 differ only in hidden interior flavor assignment while preserving those witnesses, no boundary-admissible test distinguishes them. Treating one as authoritative would require a selector over witness-equivalent assignments, hence E1, or a carrier authorizing hidden reference, hence E3. If no such selector/carrier is introduced, the distinction is bookkeeping-inert. Therefore hidden transported flavor preserving all PMNS data collapses under admissible witness fixation. $\square$

7 AASC closure spine translated into the neutrino domain

The closure upgrade uses AASC as a constraint architecture, not as a new neutrino dynamics. AASC does not add particles, Hamiltonian terms, measurement rules, or oscillation parameters. It specifies the conditions under which a claimed construction remains coherent as the same evaluated object on a fixed domain. The relevant corpus architecture is already present in the supplied AASC ZIP and is imported in the following physics-facing form.

Theorem 7.1 (Kernel instantiation for the neutrino construction). *Let $\text{CC}(\mathcal{D}_\nu)$ assert that a neutrino source trace, declared propagation interval, and detector trace are being evaluated as one coherent same-domain construction of a transition witness. Then the AASC kernel*

$$K_{\text{AASC}} = \{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\} \quad (40)$$

is instantiated on $\mathcal{D}_\nu$ as a necessity for that construction. In the neutrino domain, this means: admitted transition witnesses must be preserved, source and detector reference must remain fixed, identity claims must be reuse-stable across licensed redescription, and later detector information may not repair an earlier source assignment while counting as the same evaluated history.

Proof. If **Ref** is absent, the source label α and detector label β do not remain fixed targets of the same history; the expression “the $\alpha \rightarrow \beta$ transition” loses determinate reference. If **Stand** is absent, a flavor assignment cannot be reused across propagation and redescription as the same identity-bearing claim; it is merely a local inscription. If **Irr** is absent, a failed assignment can be repaired after detection while being counted as the same act, eliminating the distinction between admissible continuation and retrospective rescue. If **Adm** is absent, there is no fixed boundary deciding whether the composed source–propagation–detector history preserves the declared witness family. Each failure destroys coherent construction of the same transition object rather than yielding a weaker version of it. Therefore the kernel is not optional load; it is instantiated by the target construction itself. $\square$

Corollary 7.2 (AASC is proof infrastructure, not extra phenomenology). *The AASC layer does not compete with PMNS dynamics. It supplies the fixed-domain constraint discipline required to decide whether an identity assignment is admissible once the PMNS propagation envelope and boundary witnesses are already declared.*

AASC closure source	Imported closure role	Neutrino translation
Minimal Conditions for Admissible Construction	Kernel non-optional; no faithful lower same-domain generator; standing is conserved rather than created or repaired [9].	A flavor identity claim must already preserve boundary fixation, propagation witness, and act-time non-repair. It cannot be added after detection.
Standing–Admissibility Identity Closure	No second same-domain classifier; divergence routes collapse into inertness, repair, hidden classifier, or standing violation [10].	A hidden flavor classifier cannot coexist with the PMNS witness as a second standing gate.

AASC closure source	Imported closure role	Neutrino translation
Boundary-Trace Fixation under Admissibility	When internal selectors are exhausted, a declared boundary trace is the admissible anchor; absence, multiplicity, instability, and post-selection fail closed [11].	Weak-interaction production and detection fix flavor only at endpoints. They do not select an interior flavor representative.
AMetric Boundary	No selector, ordering, metric, probability, typicality rule, canonical representative, or discrete index has boundary authority unless already fixed as witness, anchor, boundary trace, or transition invariant [15].	No hidden branch, preferred basis representative, or classical mass/flavor index may authorize fixed flavor standing at the propagation interior.
Fixed-Domain Exhaustion Theorem	Same-scope admissibility-bearing variation must alter boundary typing, standing partition, lawful quotient, or admissible continuation; no fifth governance-bearing locus remains [13].	A rival must alter flavor boundaries, transition probabilities, propagation law, or scope. If it alters none, it is bookkeeping.
Closure by Exhaustion for Same-Scope Operators	Same-scope operators are bookkeeping-equivalent or inadmissible; totalization, repair, hidden data, foundation swap, and transfer do not create a new admissible family [14].	A fixed-label transport operator cannot rescue conserved flavor while preserving the same PMNS witness.
Anchor, Tensor, and Skin	Every admissibility-bearing descriptive element is uniquely typed as anchor, tensor, or skin; no fourth role survives [16].	Boundary flavor traces are anchors; H , U , phases, and amplitudes are tensor/witness content; basis notation is skin.
Standard-Model Classification	Same-regime Standard-Model bookkeeping structures are unique up to admissible equivalence; BSM proposals are not same-scope counterexamples unless they preserve the same witness and boundary ledger [17].	PMNS coordinates are admitted Standard-Model witness/envelope data, not an independent flavor ontology. BSM additions are scope changes unless witness-equivalent.

The terms used below are therefore translated as follows.

AASC term	Physics-facing use in this manuscript
Kernel	The instantiated constraint package $\{\text{Adm}, \text{Stand}, \text{Ref}, \text{lrr}\}$ required for coherent same-domain construction.
Standing	Reuse-stable, witness-preserving identity status across the declared history.
Admissibility	Preservation of the evaluable transition structure on the fixed domain.
Boundary trace	A weak-interaction event record fixing flavor at production or detection.
AMetric boundary	The pre-selection interface where no hidden selector or canonical representative has authority unless already fixed by the declared domain.
Anchor	Endpoint data required for admissible reference: here, flavor production/detection traces.
Tensor	Propagation and quotient-level invariant content: here, Hamiltonian, mixing, phases, amplitudes, and transition witnesses.
Skin	Representation surplus: basis notation, parameterization, equivalent charts, and verbal relabeling.
Same-scope exhaustion	No non-bookkeeping rescue exists unless it changes boundary maps, witness family, propagation law, role typing, or envelope.

Principle 7.3 (Imported AASC closure schema). On a fixed domain with committed boundary traces, admissible transition witnesses, and identity-preserving redescription, a candidate identity carrier is admissible only if it is boundary-fixed, transport-invariant, and witness-preserving. If it is not boundary-fixed, it is a selector. If it is not transport-invariant, it is not an identity carrier. If it is not witness-preserving, it changes the domain. If it is witness-preserving but adds no standing-bearing distinction, it is bookkeeping.

This schema is the closure bridge from “fixed flavor fails” to “the oscillation kernel is the unique same-scope reconstruction class up to admissible bookkeeping equivalence.”

8 Flavor as boundary-trace fixation

Definition 8.1 (Neutrino boundary-propagation domain). A neutrino boundary-propagation domain is

$$\mathcal{D}_\nu = (\mathcal{H}_\nu, \mathcal{F}, H, \mathcal{U}, \text{Tr}_E, \text{Tr}_D, \mathcal{W}, \mathcal{S}, \mathcal{E}), \quad (41)$$

where $\mathcal{F} = \{e, \mu, \tau\}$ is the weak flavor alphabet; Tr_E and Tr_D are emission and detection trace maps; $\mathcal{W}$ contains the admitted transition amplitudes or probabilities; $\mathcal{S}$ records approximation class, baselines, energies, matter profiles, and coherence assumptions; and $\mathcal{E}$ records numerical inputs such as mixing parameters and mass splittings.

Definition 8.2 (Boundary flavor trace). A boundary flavor trace is a weak-interaction event record. At production, $\text{Tr}_E = \alpha$ means that the charged-current source event fixes the source label α . At

detection, $\text{Tr}_D = \beta$ means that the detector event fixes the sink label β . These are endpoint anchors. They are not automatically propagation invariants.

The witness-complete status of this boundary algebra is now fixed formally by Theorem 5.6. The present section therefore uses weak-interaction traces only in their boundary role: they fix endpoint flavor labels but do not license a propagation-invariant flavor assignment.

Definition 8.3 (Propagation-invariant flavor assignment). A flavor assignment α is propagation-invariant on $\mathcal{D}_\nu$ if the corresponding flavor projector is invariant under the declared propagation family:

$$\mathcal{U}(L)^\dagger P_\alpha \mathcal{U}(L) = P_\alpha \quad \forall L \in \mathcal{S}. \quad (42)$$

When this holds, flavor can be treated as a standing-bearing identity through the propagation interval. When it fails, flavor can still be an endpoint trace but not an interior conserved identity.

Theorem 8.4 (Boundary trace does not imply interior identity). *A weak-interaction flavor trace at emission or detection fixes a boundary anchor, not an interior identity carrier. It becomes an interior standing-bearing identity only if an admissible transport invariant carries the same label through the declared propagation family.*

Proof. Boundary-trace fixation supplies endpoint reference. It does not, by itself, supply a propagation map. To reuse the label as an identity across the interior, one must preserve it under the licensed transition structure and under redescription of the same physical history. In the neutrino domain, that condition is precisely invariance of the flavor projector under $\mathcal{U}$. Without this invariant, the label remains fixed at the boundary but cannot be moved into the interior without adding a selector, changing the witness, or repairing the history after the fact. Those moves are excluded by boundary-trace fixation, AMetric no-selector discipline, and same-domain non-repair. $\square$

9 Propagation-interaction noncommutation

Lemma 9.1 (Flavor invariance iff commutation). *Let $\mathcal{U}(t) = e^{-iHt}$ be a strongly continuous unitary propagation family and let P_α be a flavor projector. Then*

$$\mathcal{U}(t)^\dagger P_\alpha \mathcal{U}(t) = P_\alpha \quad \text{for all } t \quad (43)$$

if and only if $[H, P_\alpha] = 0$ on the relevant domain.

Proof. If $[H, P_\alpha] = 0$, then P_α commutes with e^{-iHt} for all t , so the projector is invariant. Conversely, if the projector is invariant for all t , differentiating $\mathcal{U}(t)^\dagger P_\alpha \mathcal{U}(t)$ at $t = 0$ gives $i[H, P_\alpha] = 0$. $\square$

Corollary 9.2 (Noncommuting propagation forbids conserved flavor). *If $[H, P_\alpha] \neq 0$ on the declared interval class, then α is not a conserved propagation identity on that domain.*

Proof. A conserved propagation identity requires invariant transport of the corresponding projector. By Lemma 9.1, invariant transport is equivalent to commutation. Noncommutation therefore blocks fixed flavor conservation. $\square$

Theorem 9.3 (Propagation–interaction noncommutation forcing). *Fix $\mathcal{D}_\nu$ with boundary flavor trace α and admitted coherent transition witness $\mathcal{W}_{\alpha\beta}$. Suppose $[H, P_\alpha] \neq 0$. Any same-scope account that nevertheless asserts fixed flavor transport must fall into one of the following dispositions:*

- (F1) *it enforces $\mathcal{U}^\dagger P_\alpha \mathcal{U} = P_\alpha$ and thereby contradicts the declared propagation witness;*
- (F2) *it preserves all witnesses while adding no standing-bearing distinction, and is bookkeeping-equivalent to the standard kernel;*
- (F3) *it imports an unfixed hidden selector, branch index, or canonical representative;*
- (F4) *it repairs an endpoint label after detection;*
- (F5) *it changes the Hamiltonian, boundary maps, coherence assumptions, witness family, or physical scope.*

No fixed flavor transport remains as a same-domain admissible identity claim.

Proof. A fixed flavor transport claim must either attach to the flavor projector P_α or not. If it attaches to P_α , then Lemma 9.1 requires commutation. The hypothesis denies commutation, so the claim either contradicts the declared witness or changes the propagation law, giving (F1) or (F5).

If the claim does not attach to P_α , then it is no longer the boundary flavor label fixed by $\text{Tr}_E = \alpha$. If the detached label has no effect on amplitudes, probabilities, detector comparisons, or lawful reuse of the event history, it is skin-level bookkeeping, giving (F2). If it affects outcomes, it supplies standing-bearing work not fixed by the endpoint trace or propagation invariant; by AMetric no-selector discipline and standing–admissibility identity closure, this is an imported hidden selector or second classifier, giving (F3). If the detached label is chosen only after the detector record is available, it is post-hoc repair, giving (F4). If the construction changes dynamics, measurement, decoherence, hidden sectors, or boundary channels, it is no longer same-scope, giving (F5). The fixed-domain exhaustion and same-scope operator closure theorems leave no sixth locus. $\square$

Remark 9.4 (Why this is stronger than basis mismatch). Basis mismatch says that the coordinate system used at the boundary differs from the coordinate system convenient for propagation. Theorem 9.3 says more: if the boundary coordinate is promoted into a conserved identity, the noncommuting propagation witness forces collapse, bookkeeping, selector import, repair, or scope change.

10 No present-only or hidden flavor transport

The operator obstruction blocks literal fixed flavor transport. A hostile reader may still posit a hidden label that rides behind the standard amplitudes. The AASC closure architecture blocks that move only if the obstruction is stated at the witness level.

Definition 10.1 (Hidden flavor transport). A hidden flavor transport is a same-scope rule assigning an internal label λ to the emitted neutrino and a transport law $T_L(\lambda)$ such that λ is claimed to preserve the source flavor identity while the ordinary boundary probabilities $P_{\alpha \rightarrow \beta}$ are retained.

Lemma 10.2 (No present-only flavor identity). *On a noncommuting neutrino domain, equality of the source boundary flavor label is not a congruence for future boundary outcomes. Therefore the present label alone cannot serve as a standing identity quotient for the committed propagation history.*

Proof. Let two histories have the same source trace α but different licensed propagation intervals, effective Hamiltonians, or phase accumulations within $\mathcal{S}$. Because $[H, P_\alpha] \neq 0$, there exist licensed continuations for which the downstream overlaps with some P_β differ. Hence the relation “same present source flavor” is not preserved under concatenation with admissible propagation and detection. By the fixed-domain no-present-only identity rule, a relation that fails congruence under irreversible committed continuation cannot serve as the standing quotient. It is an endpoint anchor, not an interior identity. $\square$

Lemma 10.3 (No hidden flavor transport). *On a witness-complete noncommuting neutrino domain, a hidden flavor transport preserving the same coherent PMNS witness is either bookkeeping-equivalent to the boundary kernel, an inadmissible hidden selector, or a scope change. It cannot be an additional same-domain conserved flavor identity.*

Proof. By Lemma 5.6, a standing-bearing flavor carrier must be fixed by the declared boundary/projector witnesses or else change scope. Let λ be the alleged hidden carrier. If λ is fixed by the weak boundary event and is identical to α , then conserved transport of λ requires conserved transport of P_α , which is excluded by Corollary 9.2. If λ is not identical to α , it was not fixed as flavor by the boundary trace. If it carries standing-bearing information into detection, it is an extra classifier or hidden selector. If it carries no standing-bearing information, it is bookkeeping.

The common mass-label version makes the same point explicitly. The coherent flavor state has density matrix

$$\rho_\alpha = |\nu_\alpha\rangle\langle\nu_\alpha| = \sum_{i,j} U_{\alpha i}^* U_{\alpha j} |\nu_i\rangle\langle\nu_j|. \quad (44)$$

A classical hidden mass label replaces this by the incoherent mixture

$$\rho_\alpha^{\text{mix}} = \sum_i |U_{\alpha i}|^2 |\nu_i\rangle\langle\nu_i|, \quad (45)$$

which removes the off-diagonal interference terms. That changes the admitted witness except in averaged or degenerate scopes. If hidden phases or amplitudes are added to restore the interference terms, the construction has simply rebuilt the ordinary amplitude kernel in hidden notation, unless it changes the domain by adding new dynamical load. Therefore no hidden label supplies an additional same-domain conserved flavor identity. $\square$

Corollary 10.4 (No mass-label substitution). *A mass eigenlabel may be a propagation tensor component. It is not a substitute for fixed weak flavor identity in a flavor-produced event. Treating it as a classical hidden identity either destroys the coherent witness or imports unlicensed load.*

11 Two-flavor witness

A two-flavor model displays the closure in a form that is hard to hide behind notation. Let boundary flavor kets $|\nu_a\rangle$ and $|\nu_b\rangle$ be related to propagation eigenkets by

$$|\nu_a\rangle = \cos\theta |\nu_1\rangle + \sin\theta |\nu_2\rangle, \quad (46)$$

$$|\nu_b\rangle = -\sin\theta |\nu_1\rangle + \cos\theta |\nu_2\rangle. \quad (47)$$

With relative phase $\Delta\phi = \Delta m^2 L/(2E)$, the coherent transition probability is

$$P_{a \rightarrow b}^{\text{coh}}(L, E) = \sin^2(2\theta) \sin^2\left(\frac{\Delta m^2 L}{4E}\right). \quad (48)$$

Three alternatives show the exhaustion.

- (W1) **Fixed flavor conservation:** $P_{a \rightarrow b}(L, E) = 0$ for all L . This contradicts Eq. (48) when the witness is nontrivial.
- (W2) **Classical hidden mass label:** replacing the coherent state by a mixture gives

$$P_{a \rightarrow b}^{\text{mix}} = \sum_i |\langle \nu_i | \nu_a \rangle|^2 |\langle \nu_b | \nu_i \rangle|^2 = \frac{1}{2} \sin^2(2\theta), \quad (49)$$

which removes the oscillatory phase witness.

- (W3) **Amplitude-preserving hidden label:** adding enough hidden structure to recover Eq. (48) reproduces the boundary kernel. If the hidden structure changes no witness, it is bookkeeping; if it changes a witness, it is new scope.

Example 11.1 (What the two-flavor witness proves). The witness does not derive θ or Δm^2 . It proves that once nonzero mixing and nondegenerate phase are admitted as envelope inputs, fixed flavor transport is unavailable. The only same-scope witness-preserving reconstruction is the amplitude kernel.

12 Boundary-to-boundary reconstruction

Definition 12.1 (Boundary-to-boundary transition kernel). For a declared propagation family $\mathcal{U}(L, E)$, define

$$K_{\beta\alpha}(L, E) = \langle \nu_\beta | \mathcal{U}(L, E) | \nu_\alpha \rangle, \quad P_{\alpha \rightarrow \beta}(L, E) = |K_{\beta\alpha}(L, E)|^2. \quad (50)$$

The matrix K is the amplitude-level boundary reconstruction kernel; P is the probability-level boundary witness.

The operator category $\mathcal{C}_{\text{PMNS}}$, its admissible morphisms, the equivalence relation $\sim_{\text{adm}}$, and the categorical reconstruction-exhaustion theorem have been fixed in Section 4. The present section identifies the concrete representative of that equivalence class: the boundary-to-boundary kernel.

Theorem 12.2 (Boundary-to-boundary reconstruction theorem). *On a witness-complete fixed neutrino boundary-propagation domain, the kernel $K_{\beta\alpha}$ is the unique same-scope object, up to admissible bookkeeping equivalence, that composes the emission flavor trace, the declared propagation family, and the detection flavor trace.*

Proof. The source boundary trace fixes $|\nu_\alpha\rangle$ as the input anchor. The propagation tensor is $\mathcal{U}(L, E)$. The detector boundary trace tests against $\langle \nu_\beta |$. Their composition is the matrix element $\langle \nu_\beta | \mathcal{U} | \nu_\alpha \rangle$. A basis change or parameterization that preserves this matrix element is skin-level redescription. A construction that changes the matrix element changes the witness family, propagation law, boundary map, or scope. A construction that adds an intermediate identity carrier not fixed by the boundary or invariant under propagation imports a selector. Thus the kernel is unique up to bookkeeping equivalence, as made explicit by Theorem 4.18. $\square$

Theorem 12.3 (Oscillation realization theorem). *Assume a finite witness-complete PMNS scope with boundary flavor traces, coherent propagation, complete flavor detection channels in the declared sector, and $[H, P_\alpha] \neq 0$ for at least one source flavor α . Then the unique same-scope admissible reconstruction class, up to bookkeeping-equivalent redescription in $\mathcal{C}_{\text{PMNS}}$, is the boundary kernel K . If the off-diagonal entries of K are nonzero on the declared interval class, the empirical realization of that reconstruction is neutrino oscillation.*

Proof. By Theorem 7.1, the source–propagation–detector history already instantiates the AASC kernel when it is treated as a coherent construction of a transition witness. By Corollary 9.2, fixed flavor transport is unavailable. By Theorem 9.3 and Lemma 10.3, fixed-label rescue, hidden transport, post-hoc repair, and selector import are unavailable in the same scope. By Lemma 5.6, there is no additional same-scope flavor witness outside the boundary projectors and transition kernel. By Theorem 4.18, any same-scope admissible reconstruction rule is bookkeeping-equivalent to K .

Noncommutation means that the evolved source state is not confined to the original flavor projector for all licensed intervals. In a witness-complete flavor detection sector, this produces nontrivial overlaps with other flavor projectors whenever coherence and phase structure allow the corresponding off-diagonal kernel entries. Those baseline/energy-dependent boundary probabilities are precisely what the standard formalism calls neutrino oscillation. Therefore oscillation is not an extra identity process added to propagation. Given coherent noncommuting propagation, it is the unique same-scope admissible reconstruction class, up to bookkeeping-equivalent redescription, once fixed flavor identity has failed. $\square$

Corollary 12.4 (Identity non-conservation without conservation-law violation). *The result does not assert violation of energy, momentum, gauge charge, or any Standard-Model conservation law. It says only that weak flavor, as a boundary label, is not a conserved standing-bearing identity under noncommuting propagation.*

13 Rival exhaustion

The following theorem is the hard closure layer. It is not a list of rhetorical objections; it is the fixed-domain image of the AASC exhaustion architecture applied to the neutrino target.

Theorem 13.1 (Rival exhaustion for fixed flavor transport). *Let $R \in \mathfrak{R}(\mathcal{D}_\nu)$ be a same-scope rival to the boundary-kernel reconstruction of flavor on a finite witness-complete noncommuting PMNS domain. Then R falls into at least one of the following classes, each of which collapses as a fixed-flavor rescue:*

- (R1) Fixed flavor conservation:** requires $[H, P_\alpha] = 0$, contrary to the target case.
- (R2) Hidden fixed flavor:** introduces a second classifier or has no witness effect.
- (R3) Hidden classical mass identity:** destroys coherent interference or becomes kernel bookkeeping.
- (R4) Basis relabeling:** preserves K and is skin-level notation.
- (R5) Collapse-only or decoherence account:** changes the coherence scope or yields averaged transition weights, not restored fixed flavor identity.

- (R6) Arbitrary boundary-dependent flavor reconstruction:** *fails witness preservation unless it equals K .*
- (R7) Parameter-tuned non-oscillation:** *describes a commuting, degenerate, or zero-mixing scope, not the noncommuting target domain.*
- (R8) Detector-only reassignment:** *changes boundary maps or repairs after detection.*
- (R9) Hidden witness sector:** *if it couples observably, it changes the witness family; if it does not, it is bookkeeping, disconnected, or a hidden selector.*
- (R10) BSM or UV completion:** *if it preserves the same witness and boundary ledger it reduces to the same kernel on this scope; if it adds fields, sectors, channels, or dynamics, it changes scope.*

No further same-scope fixed-flavor competitor remains.

Proof. A rival to K must differ at one of the AASC local loci: boundary typing, standing partition, lawful quotient/transport, continuation history, role typing, witness family, or envelope. If it differs at none, it is bookkeeping-equivalent to K . If it changes boundary typing, it is (R8) or a scope change. If it changes the standing partition by adding fixed flavor, it is (R1) or (R2). If it changes lawful transport, it is (R3), (R5), or (R10). If it is merely a different coordinate skin, it is (R4). If it changes the physical regime so that noncommutation or coherence no longer holds, it is (R5) or (R7). If it supplies arbitrary reconstruction not determined by the declared witness, it is (R6). If it adds an additional witness sector, it is (R9) by Corollary 5.7. These loci are exhaustive by fixed-domain exhaustion, same-scope operator closure, and the reconstruction category in Section 4. Hence no same-scope fixed-flavor competitor survives. $\square$

14 Relation to standard neutrino physics

Why AASC is needed. The AASC machinery is not used to redescribe oscillation phenomenology. The ordinary Hilbert-space calculation already does that. AASC is used as a constraint formalism: it supplies the comparison-class discipline needed to distinguish admissible reconstruction from hidden selector insertion, scope extension, post-hoc relabeling, and redundant notation. Once the kernel is instantiated by coherent construction, a proposed flavor carrier must be boundary-fixed, transport-invariant, witness-preserving, non-repairing, and free of hidden selectors. Those are exactly the constraints a fixed-flavor rescue fails.

PMNS formalism. The PMNS formalism remains untouched. The present theorem does not infer the values of $U_{\alpha i}$, Δm_{ij}^2 , CP-violating phases, or absolute masses. It classifies the identity status of weak flavor once those quantities are admitted as envelope inputs.

Matter effects. Matter effects alter the effective Hamiltonian. The theorem applies to the Hamiltonian declared by the scope. The question remains whether the relevant flavor projectors commute with that Hamiltonian and whether the same boundary witness is preserved.

Dirac/Majorana status. The obstruction is insensitive to the Dirac/Majorana distinction for ordinary flavor oscillation probabilities. Majorana-sector questions may require different witnesses, but they do not restore fixed weak flavor transport on the oscillation scope.

Charged leptons. Charged leptons do not show analogous ordinary vacuum flavor oscillation

because the relevant interaction labels and free propagation labels are aligned in the declared scope. Their identity projectors commute with the propagation Hamiltonian to the accuracy and role under discussion. Neutrinos are special because weak flavor is endpoint-fixed while propagation is governed by mass or effective matter eigenstructure.

Wave packets and decoherence. Wave-packet separation and decoherence can suppress or average oscillatory interference. That changes the witness regime from coherent oscillation to averaged transition weights. It does not revive fixed flavor identity in the original coherent noncommuting scope.

Interpretation. The theorem is neither flavor anti-realism nor hidden ontology. Flavor is real as an interaction-defined boundary label. Propagation structure is real as the declared tensor witness. What is inadmissible is the further claim that weak flavor is a globally transported identity when the declared operator does not preserve it.

15 AASC dependency ledger

Target step	AASC closure used	Manuscript location
AASC kernel instantiated by coherent source–propagation– detector construction	Minimal Conditions for Admissible Construction; coherent evaluability architecture	Theorem 7.1
Flavor fixed only at interaction endpoints	Boundary-trace fixation; anchor role discipline	Theorem 8.4
Fixed flavor requires transport invariance	Standing as reuse-stable admissibility; identity-preserving redescription	Lemma 9.1, Corollary 9.2
Basis mismatch becomes an obstruction only when a conserved-identity claim is made	Minimal admissible construction; witness-preserving transition continuity	Theorem 9.3
Flavor witnesses complete on declared scope	Detector-coupled witness completion; Standard-Model classification; standing/admissibility identity closure	Theorem 5.6
Finite PMNS comparison class locked	Fixed-domain exhaustion; same-scope operator closure; finite witness-complete PMNS scope	Theorem 2.3
Operator comparison class made explicit	PMNS reconstruction category; morphism classification; admissible equivalence	Section 4, Theorem 4.17

Target step	AASC closure used	Manuscript location
Hidden witness sector collapse	Standing–admissibility identity closure; AMetric no-selector boundary; quantum boundary compatibility	Corollary 5.7
No hidden flavor classifier	Standing–admissibility identity closure; AMetric no-selector boundary	Lemma 10.3
No present-only flavor identity	Fixed-domain exhaustion; no present-only identity under irreversible composition	Lemma 10.2
No same-scope rescue operator	Closure by exhaustion for same-scope operators	Theorem 13.1
PMNS parameters remain admitted inputs	Standard-Model bookkeeping classification; envelope boundary ledger	Sections 3 and 14
Reconstruction comparison category fixed and exhausted	Fixed-domain exhaustion; anchor/tensor/skin; same-scope operator closure	Definitions 4.1–4.3, Theorem 4.18
Oscillation as unique equivalence-class reconstruction	Boundary-trace fixation + operator exhaustion + fixed-domain exhaustion + SM classification	Theorem 12.3
BSM non-counterexample discipline	Standard-Model classification and BSM non-competition theorem	Theorem 13.1

16 Hostile-referee attack ledger

Attack	Closure response
“This does not derive oscillation.”	Correct. It is not a numerical derivation. It is a same-scope classification theorem: fixed flavor transport is not an admissible identity claim once noncommuting propagation and boundary traces are declared.
“This is just textbook basis mismatch.”	Basis mismatch alone is representation. The closure result adds boundary fixation, standing identity, and same-scope exhaustion. It proves that a conserved flavor identity cannot be added to the textbook formalism.
“Flavor is merely conventional.”	Boundary flavor is fixed by weak interaction traces. The rejected object is not flavor at endpoints but globally fixed flavor transport through the interior.

Attack	Closure response
“A hidden label carries the real identity.”	If it changes no witness, it is bookkeeping. If it changes a witness, it changes scope. If it is not boundary-fixed or transport-invariant, it is an inadmissible selector.
“The real identity is mass.”	Mass labels are propagation tensor components. Treating them as classical hidden identities for a flavor-produced event destroys interference or imports hidden load.
“Matter effects change the story.”	Matter changes the declared H . The theorem applies to the effective Hamiltonian in the declared matter envelope.
“Decoherence removes oscillation.”	Decoherence changes the witness regime. It can average the kernel; it does not restore fixed flavor identity in the coherent noncommuting scope.
“BSM physics may explain oscillation differently.”	A BSM proposal preserving the same witness reduces to the same boundary kernel on this scope. A proposal adding new fields, channels, or dynamics is larger-domain physics, not a same-scope fixed-flavor rescue.
“The comparison class is undefined.”	The reconstruction class is defined by six same-scope constraints: anchors, tensor, witnesses, skin covariance, no hidden selector, and no repair. The exhaustion theorem proves every admissible member reduces to K .
“AASC is extra machinery.”	AASC is proof infrastructure for coherent construction. The kernel is instantiated by the target history itself; it adds no dynamics or parameters.
“This is semantic.”	The result forbids a class of identity assignments and same-scope rescue operators. It is a structural constraint on admissible descriptions, not a vocabulary preference.
“The theorem is anti-realist.”	It is not. It preserves boundary flavor, propagation structure, and oscillation data. It rejects only propagation-invariant weak flavor assignment when the operator fails to preserve flavor projectors.

17 Conclusion

The standard PMNS transition kernel already contains the correct oscillation mathematics. The closure question is what that mathematics permits one to say about identity. A weak-interaction source event fixes flavor at the boundary. A detection event fixes flavor at another boundary. Between those events, the declared propagation operator generally fails to preserve the flavor projectors. Therefore fixed weak flavor is not an admissible interior identity.

The AASC closure architecture sharpens the result from obstruction to equivalence-class uniqueness

because the kernel is instantiated as a necessity for coherent construction, not stipulated as an optional interpretation. Hidden fixed labels, present-only identity, basis relabeling, classical mass labels, post-hoc detector repair, arbitrary reconstruction, and same-scope operator rescues all collapse into bookkeeping, witness failure, hidden selector import, or scope change. The unique same-scope admissible equivalence class in $\mathcal{C}_{\text{PMNS}}$ is represented by the boundary-to-boundary kernel $K_{\beta\alpha} = \langle \nu_\beta | \mathcal{U} | \nu_\alpha \rangle$. In the coherent nontrivial case, that kernel is observed as neutrino oscillation.

Thus neutrino oscillation is not merely compatible with admissible propagation. On the declared finite witness-complete noncommuting PMNS domain, it is the unique admissible reconstruction class of weak flavor, up to bookkeeping-equivalent redescription, after fixed flavor transport has been closed. The AQFT embedding states the same result as a boundary-net reconstruction sector: flavor witnesses live at localized source and detector boundaries, while the propagation interior supplies coherent evolution but no admissibly witnessed weak-flavor selector. The appendix remains a boundary-net embedding rather than a full constructive AQFT model, but it is sufficient to show that the reconstruction claim can be read algebraically as preservation of boundary transition functionals. The final operator-domain obstruction strengthens the endpoint: globally transported flavor is not merely unselected; it is an inadmissible totalization of a boundary-defined partial operator inside $\mathcal{C}_{\text{PMNS}}^\partial$. The nonvanishing internal admissible-totalization obstruction $\mathfrak{D}_{\text{flavor}}^\partial$ records this failure, and the universal quotient $\mathcal{Q}_{\text{PMNS}}^\partial$ states the final uniqueness result: every admissible same-scope reconstruction factors through the oscillatory boundary-kernel class up to bookkeeping equivalence.

A AQFT Embedding: Boundary Nets and Reconstruction Sectors

This appendix does not construct a full Haag–Kastler model of neutrino production and detection, and it does not assume net completion, covariance, isotony, locality, or representation-theoretic hypotheses beyond the schematic boundary-net notation explicitly displayed here. No DHR-style superselection classification or Haag–Kastler construction is claimed. The discussion is interpretive and organizational: it records the localized-algebraic reading of the finite PMNS reconstruction theorem. Its purpose is to show that the argument is not tied to informal Hilbert-space language: it can be expressed as boundary witness reconstruction over local algebras.

Let the relevant spacetime regions be an emission boundary region O_{in} , a propagation tube or interior region O_{prop} , and a detection boundary region O_{out} . Associate a local net

$$O \longmapsto \mathcal{A}(O), \quad (51)$$

with local algebras

$$\mathcal{A}_{\text{in}} = \mathcal{A}(O_{\text{in}}), \quad \mathcal{A}_{\text{prop}} = \mathcal{A}(O_{\text{prop}}), \quad \mathcal{A}_{\text{out}} = \mathcal{A}(O_{\text{out}}). \quad (52)$$

Flavor witnesses are boundary projections:

$$P_\alpha^{\text{in}} \in \mathcal{A}_{\text{in}}, \quad P_\beta^{\text{out}} \in \mathcal{A}_{\text{out}}. \quad (53)$$

The propagation interior carries the declared coherent evolution. It does not carry an admissibly fixed flavor selector

$$S_\alpha \notin \mathcal{A}_{\text{prop}} \quad (54)$$

unless that selector is definable from the boundary witness data and preserves the boundary transition functionals.

Let ω be a state and let α_t denote the declared time-evolution automorphism. The boundary-to-boundary transition functional may be written schematically as

$$\omega_{\alpha \rightarrow \beta}(t) = \omega\left(P_{\beta}^{\text{out}} \alpha_t(P_{\alpha}^{\text{in}}) P_{\beta}^{\text{out}}\right), \quad (55)$$

or, in the finite Hilbert-space reduction used in the main text,

$$\omega_{\alpha \rightarrow \beta}(t) = \omega(P_{\beta} \alpha_t(P_{\alpha})). \quad (56)$$

Definition A.1 (Boundary representation of the finite PMNS net). A boundary representation of the finite PMNS net is a representation ρ of the displayed boundary-net data on a Hilbert space $\mathcal{K}_{\rho}$ such that the source projections $\rho(P_{\alpha}^{\text{in}})$, detector projections $\rho(P_{\beta}^{\text{out}})$, and evolution $\rho(\alpha_t)$ generate the same boundary transition functionals as the finite PMNS witness family. Two such representations are admissibly equivalent when all functionals $\omega_{\alpha \rightarrow \beta}(t)$ agree for the declared boundary alphabet and propagation intervals.

Definition A.2 (Boundary-functional reconstruction sector). A boundary-functional reconstruction sector is an equivalence class

$$[\rho]_{\partial} := \{\rho' : \omega_{\alpha \rightarrow \beta}^{\rho'}(t) = \omega_{\alpha \rightarrow \beta}^{\rho}(t) \text{ for all admitted } \alpha, \beta, t\} \quad (57)$$

under boundary-functional equivalence. This is a boundary-functional reconstruction-sector notion, not a DHR or superselection-sector classification. It records equality of the finite PMNS boundary witness representation.

Theorem A.3 (Boundary-functional sector form of PMNS reconstruction). *The oscillatory PMNS reconstruction determines a unique boundary-functional reconstruction sector $[\rho_K]_{\partial}$, where ρ_K is any boundary representation whose transition functionals realize the kernel $K_{\beta\alpha}$. A representation with a globally transported weak-flavor selector either changes the boundary functionals, is equivalent to $[\rho_K]_{\partial}$, or exits the declared finite PMNS representation scope.*

Proof. The kernel $K_{\beta\alpha}$ fixes the boundary functionals and therefore determines the sector $[\rho_K]_{\partial}$. If another representation preserves all such functionals, it lies in the same sector by definition. If it changes one, it represents a different witness family. If it adds a propagation-local flavor selector while preserving all boundary functionals, the selector contributes no admissible boundary-functional distinction and collapses to bookkeeping; if it is made standing-bearing, it is an added witness sector or a hidden selector and therefore exits the declared representation scope. This is the boundary-functional sector reading of Corollary 4.15 and Theorem 6.6. $\square$

Definition A.4 (Oscillatory reconstruction sector). An oscillatory reconstruction sector is the admissible equivalence class

$$[\omega, \alpha_t, \{P_{\alpha}\}]_{\text{adm}} \quad (58)$$

under transformations preserving all boundary transition functionals of the form (55). This sector records boundary-witness preservation, not a propagation-carried flavor ontology.

Theorem A.5 (Oscillatory reconstruction sector). *In the boundary-net reading, the finite PMNS oscillatory reconstruction is a sector of boundary functional equivalence. That is, two localized descriptions represent the same oscillatory sector exactly when they preserve all functionals*

$$\omega_{\alpha \rightarrow \beta}(t) = \omega\left(P_{\beta}^{\text{out}} \alpha_t(P_{\alpha}^{\text{in}}) P_{\beta}^{\text{out}}\right) \quad (59)$$

for every admitted source channel, detector channel, and propagation interval. A description that changes these functionals changes the sector; a description that preserves them is the same sector up to admissible bookkeeping equivalence.

Proof. The displayed boundary functionals are precisely the AQFT-style representatives of the witness family $\mathfrak{F}(\mathfrak{R})$. Preservation of all of them is therefore the localized-algebraic version of transition-witness preservation, interference preservation, and basis covariance in $\mathcal{C}_{\text{PMNS}}$. If a proposed localized description changes any such functional, the detector-observable transition structure changes and the sector is different. If it preserves all such functionals, then it induces the same object of $\mathcal{W}_{\text{PMNS}}$ and hence the same admissible sector under $\sim_{\text{adm}}$. $\square$

Theorem A.6 (AQFT boundary-net embedding). *Within the finite PMNS boundary net, oscillatory flavor reconstruction corresponds to a reconstruction sector determined by boundary flavor projections and coherent interior evolution. A globally transported flavor selector would require an additional propagation-local projection family preserving the boundary functionals; such a family either commutes trivially with the declared evolution, changes the reconstruction sector, is bookkeeping-inert, or is not admissibly witnessed.*

Proof. The boundary algebras encode the source and detector flavor witnesses. The interior algebra carries propagation, not boundary flavor fixation. Suppose a propagation-local flavor projection family is added. If it is transported from the boundary projectors and commutes with the evolution, it describes a trivial or degenerate commuting case, not the nontrivial PMNS target. If it changes the boundary functionals, it changes the reconstruction sector. If it preserves all boundary functionals while adding no detectable distinction, it is bookkeeping-inert. If it is neither boundary-definable nor functionally witnessed, it is an inadmissible selector. Hence the finite PMNS oscillation theorem embeds as a boundary-to-boundary reconstruction-sector statement. $\square$

This localized reading reinforces the main text: flavor witnesses are fixed at source and detector regions, while coherent propagation supplies the interior tensor. The theorem excludes a carried weak-flavor selector inside the declared reconstruction sector; it does not deny local detector events, transition functionals, or the ordinary oscillation data.

B Neutrino Reconstruction Degeneracy Matrix

This appendix records the compact hostile-referee audit induced by the operator-domain obstruction pass. Each failure mode is blocked internally to the declared finite PMNS reconstruction category.

ID	Degeneracy / escape route	Blocking result
N1	Globally transported weak-flavor identity through the propagation interior	Flavor transport totalization obstruction, Theorem 6.3
N2	Propagation-local hidden flavor selector	No-selector condition in $\mathcal{C}_{\text{PMNS}}$; obstruction class E1, Theorem 6.8
N3	Detector-independent flavor witness inside finite PMNS scope	Weak-interaction flavor witness completion, Theorem 5.6
N4	Conservative hidden transported-flavor sector preserving all PMNS witnesses	Conservative reconstruction collapse, Theorem 6.7
N5	Retrodictive present-only flavor identity imposed at detection	No-repair history condition and no present-only transport theorem
N6	Fixed flavor conservation despite noncommuting propagation	Propagation–interaction noncommutation theorem and reconstruction exhaustion
N7	Basis relabeling treated as rival ontology	Basis covariance and admissible equivalence in $\mathcal{C}_{\text{PMNS}}$
N8	Probability, typicality, symmetry, or explanatory compression used to authorize transported flavor	Carrier laundering class E3, Theorem 6.8
N9	Added sterile/hidden sector presented as same PMNS reconstruction problem	Scope transport class E4; finite PMNS scope lock
N10	Boundary-independent flavor assignment claimed without detector coupling	Minimal boundary witness algebra, Theorem 5.5
N11	Hidden assignment preserving all data but reversing interior reference fixation	Canonical witness collapse, Corollary 6.9
N12	Treating undefined interior flavor as an error to be repaired rather than non-membership in operator domain	Undefined-is-not-error discipline, Remark 6.2; E2 obstruction

C Symbol ledger

Symbol	Meaning
$ \nu_\alpha\rangle$	Weak-interaction flavor ket fixed operationally at boundary events.
$ \nu_i\rangle$	Propagation mass eigenket in the vacuum approximation.
$U_{\alpha i}$	PMNS coordinate relating flavor and mass bases.
P_α	Flavor projector $ \nu_\alpha\rangle\langle\nu_\alpha $.
H	Declared propagation Hamiltonian or effective Hamiltonian.
$\mathcal{U}(L, E)$	Propagation family generated by H .
Tr_E, Tr_D	Emission and detection boundary trace maps.
$K_{\beta\alpha}$	Boundary-to-boundary amplitude kernel.

Symbol	Meaning
$\mathcal{W}$	Admitted transition witness family, such as oscillation amplitudes or probabilities.
$\mathfrak{R}(\mathcal{D}_\nu)$	Same-scope admissible reconstruction-class for the declared neutrino domain.
K_{AASC}	AASC kernel $\{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\}$ instantiated by coherent construction.
$\mathcal{S}$	Scope ledger: approximations, baseline/energy class, matter envelope, and coherence assumptions.
$\mathcal{E}$	Envelope ledger: numerical inputs not derived in this theorem.
Stand	Reuse-stable admissibility-bearing identity status.
AMETRIC	No-selector boundary discipline: no hidden index, metric, ordering, or representative has authority unless fixed by the declared domain.

D Proof dependency graph

- D1.** Kernel instantiation supplies admissibility, standing, reference, and irreversibility as necessary constraints on the coherent transition construction.
- D2.** Boundary-trace fixation supplies endpoint flavor anchors.
- D3.** Lemma 9.1 converts fixed flavor transport into a commutation condition.
- D4.** Corollary 9.2 blocks fixed flavor conservation in the target case.
- D5.** Theorem 9.3 exhausts direct fixed-transport rescues.
- D6.** Lemma 10.2 blocks endpoint-only identity quotients.
- D7.** Lemma 10.3 blocks hidden flavor or mass-label transport.
- D8.** Lemma 5.6 fixes the declared flavor witness family.
- D9.** Theorem 2.3 locks the finite PMNS comparison class.
- D10.** Corollary 5.7 closes hidden flavor witness sectors.
- D11.** Theorem 4.18 defines and exhausts the admissible reconstruction class.
- D12.** Theorem 12.2 identifies the unique witness-preserving composition up to bookkeeping equivalence.
- D13.** Theorem 12.3 combines the obstruction and exhaustion into unique equivalence-class reconstruction.
- D14.** Theorem 13.1 records the closed hostile-referee rival map.

E AASC source scan summary

The supplied AASC ZIP was scanned as the closure source for this upgrade and for the present mathematical-physics formalization pass. The following architectural roles were used directly:

- *Minimal Conditions for Admissible Construction*: kernel non-optionality, no faithful lower generator, act-time non-repair, standing conservation, and the claim that admissibility/standing/reference/irreversibility are instantiated by coherent construction rather than added as optional axioms.
- *Standing-Admissibility Identity Closure*: no second same-domain classifier and finite route collapse for admissibility/standing divergence.
- *Boundary-Trace Fixation under Admissibility*: endpoint traces as role anchors and failure of delayed, scalarized, unstable, or post-selected fixation.
- *Constraint Architectures for Coherent Evaluability*: witness extraction, transport covariance, no hidden evaluative remainder, fail-closed standing loss, and scope-fixed comparison discipline for evaluating a target construction.
- *A Fixed-Domain Exhaustion Theorem*: same-scope comparison discipline, no present-only identity under irreversible composition, and no fifth governance-bearing locus.
- *Closure by Exhaustion for Same-Scope Operators under Admissibility*: totalization, repair, hidden-data, diagonalization, transfer, and representation detours collapse to bookkeeping or inadmissibility.
- *The AMetric Boundary*: no selector, no ranking, no canonical representative, no typicality rule, and no discrete index at the admissibility boundary unless already fixed by the evaluated domain.
- *Anchor, Tensor, and Skin*: endpoint anchors, tensor-level witness content, and skin-level redescription as the unique admissible role decomposition.
- *A Classification Theorem for Admissible Standard-Model Structures*: PMNS and remaining numerical inputs as Standard-Model witness/envelope data, with BSM proposals treated as larger-domain unless witness-equivalent.
- *Quantum Entanglement as Boundary-Level Compatibility and Admissible Record Construction in Physically Non-Selective Quantum Regimes*: no hidden common carrier or non-selective physical regime supplies an unlicensed same-scope witness sector.
- *Partial Algebras, Operator Domains, and the Impossibility of Total Extension*: partial operators encode admissibility intrinsically through their domains; admissible extensions preserve domains on the original carrier and cannot totalize an old partial operator.
- *Closure by Exhaustion of Standing-Bearing Operator Regimes*: conservative extensions collapse to definitional bookkeeping; syntactic coverage expansion exhausts into selector insertion, repair/generator, carrier laundering, or scope transport.
- *A Canonical Witness of Reference-Dependent Confirmation*: operational equivalence with reference-dependent divergence supplies the witness pattern used to collapse hidden transported flavor assignments preserving all PMNS data.
- *Apex Standalone Closure Theorem*: no discrete multiplicity, no repairs, no generators, no carriers, and uniqueness of the admissible interior support the universal quotient construction.

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